

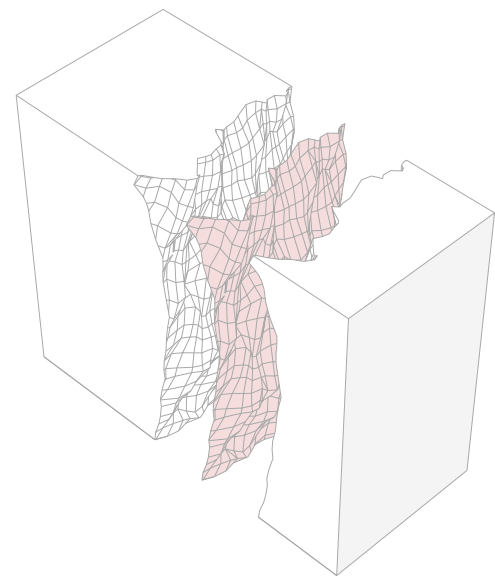


libharmonics : A C++ Library for Morphological Analysis and Spectral Galerkin Methods

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Introduction

A modern C++ library for morphological analysis and spectral Galerkin methods on parametric surfaces.

Motivation

This theme faithfully reproduces the visual identity of the original EPFL LaTeX Beamer class. All colours, spacing, and layout elements are matched.

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Why Typst?

Typst offers a modern alternative to LaTeX for academic slides and documents:

1. Fast compilation
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Theorem & Definition Blocks

Theorem.

Let $f : X \rightarrow \mathbb{R}$ be a continuously differentiable function on a compact manifold $X \subset \mathbb{R}^d$. Then the spectral Galerkin approximation f_N converges to f in $L^2(X)$ as $N \rightarrow \infty$.

Definition.

The **morphological distance** $d_{M(x,y)}$ between two points on a parametric surface $\mathcal{S}$ is the geodesic length of the shortest path constrained to lie on $\mathcal{S}$.

Example & Note Blocks

Example.

Consider a spherical harmonic expansion of order $N = 64$. The coefficient matrix A_n^m has $(N + 1)^2 = 4225$ entries, each computed via numerical quadrature on the unit sphere.

Note.

In practice, the coefficient matrices are highly sparse for smooth surfaces. Exploiting this sparsity reduces the computational cost from $\mathcal{O}(N^4)$ to $\mathcal{O}(N^2 \log N)$.

Alert & Proof Blocks

⚠ Warning.

The convergence rate degrades sharply when the surface contains C^0 discontinuities (creases). In such regions, spectral methods require local refinement to maintain accuracy.

Proof.

By the Riesz representation theorem, there exists a unique $u \in H_0^1(\Omega)$ satisfying the weak formulation for any $f \in L^2(\Omega)$. The Galerkin projection onto $V_N \subset H_0^1$ inherits this well-posedness.



Quote Block

The quote-block highlights key statements from literature or emphasises take-home messages:

The fundamental law of numerical analysis: the error in the discrete solution is bounded by the error in the best approximation from the discrete subspace.

— G. Strang, 1986

Key insight: computation scales linearly with input size.

Progress Bar Styles

Four built-in styles — set globally or per-slide via `config-store(progress-bar: "style")`:

"line" (default)	Thin red bar that fills left-to-right.
"dots"	Dot indicators on a dark strip — one per slide, current slide highlighted.
"fraction"	Compact text "5 / 20" on a dark strip aligned right.
"sections"	Segmented bar — each section gets its own colour block.
"none"	Disables the progress indicator entirely.

Dots Example

(this slide uses progress-bar: "dots")

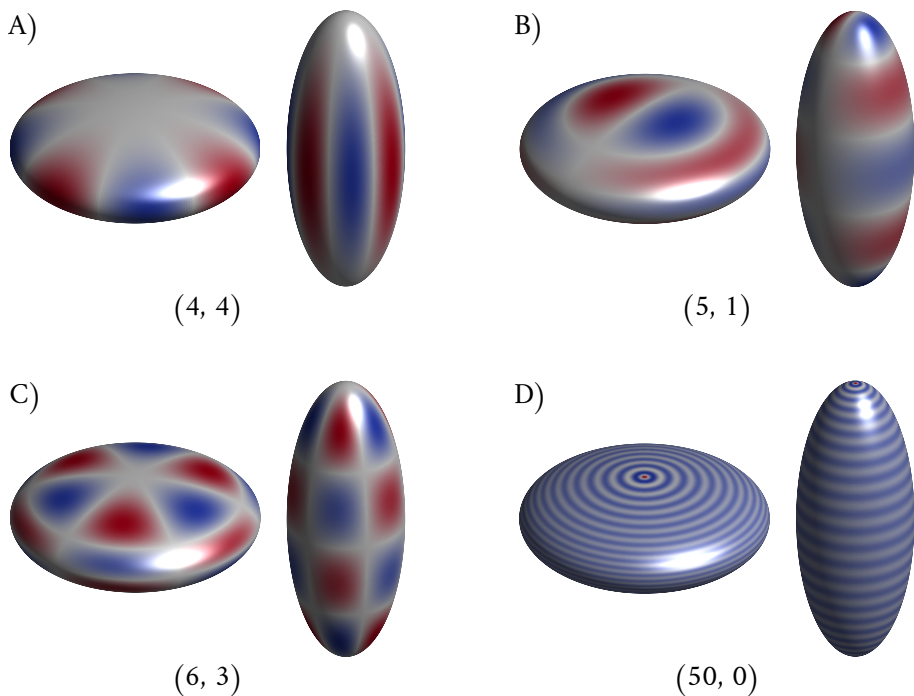
Each dot represents one slide. The current slide is highlighted in red, visited slides are light gray, upcoming are dark.

Fraction Example

(this slide uses progress-bar: "fraction")

The right side of the progress bar strip shows the current slide number over the total, e.g. “18 / 45”.

Styled Figure Captions



$$\begin{pmatrix} x(\eta, \varphi) \\ y(\eta, \varphi) \\ z(\eta, \varphi) \end{pmatrix} = \sum_{n=0}^{n_{\max}} \sum_{m=-n}^n A_n^m S_n^m(\eta, \varphi).$$

Captions are automatically styled with the theme's primary red colour and bold weight for the figure label.

Figure 1: Oblate and prolate spheroidal harmonics basis functions.

Styled Table of Contents

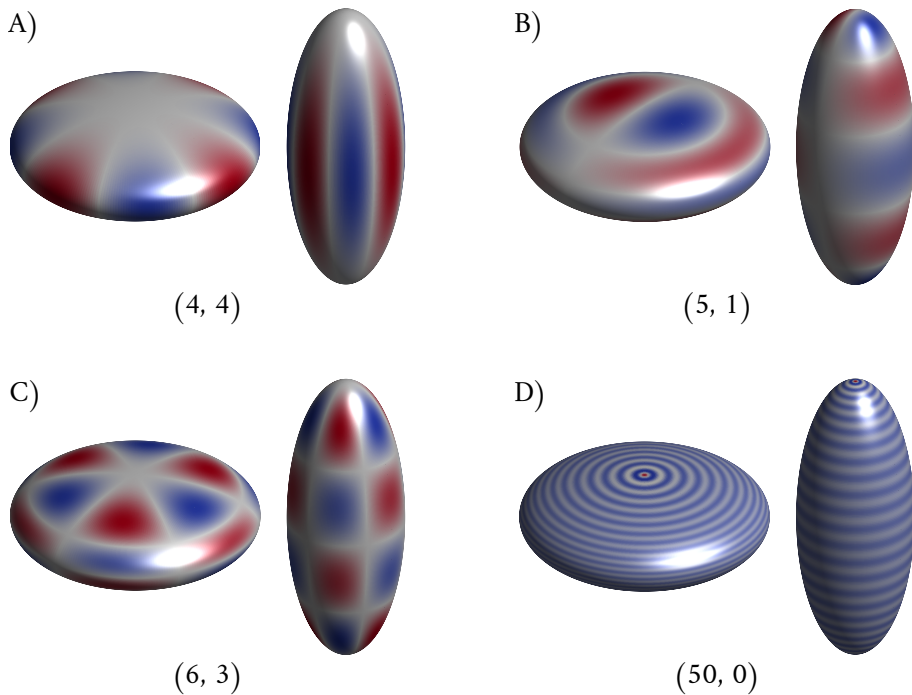
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Methodology

Spectral parameterization



$$\begin{pmatrix} x(\eta, \varphi) \\ y(\eta, \varphi) \\ z(\eta, \varphi) \end{pmatrix} = \sum_{n=0}^{n_{\max}} \sum_{m=-n}^n A_n^m S_n^m(\eta, \varphi).$$

This spectral expansion maps the surface coordinates (x, y, z) onto a basis of spherical harmonics S_n^m , weighted by coefficients A_n^m .

Figure 2: Oblate and prolate spheroidal harmonics basis functions.

Step-by-step Reveal

1. First point is always visible
- 2.
- 3.

Step-by-step Reveal

1. First point is always visible
2. Second point appears on click
- 3.

Step-by-step Reveal

1. First point is always visible
2. Second point appears on click
3. Third point appears on next click

Note: This conclusion only appears with the third point.

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+ First point is always visible
#pause
+ Second point appears on click
#pause
+ Third point appears on next click

#uncover("3-") [
  *Note:* appears with third point
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Pillar One: Theoretical Framework

The theoretical framework extends established results to handle domain-specific complexities.

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Pillar Two: Computational Methods

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Key insight: computation scales linearly with input size.

Results

Discussion

- **+25 %** accuracy improvement
- **3×** faster than prior art
- Robust across varied conditions

Conclusion

Conclusion

Conclusion

This work presents a comprehensive framework combining theoretical rigour with practical efficiency.

Future work will explore extensions to higher-dimensional problems and real-time applications.

References

- [1] M. Shaqfa and W. M. van Rees, "Spheroidal harmonics for generalizing the morphological decomposition of closed parametric surfaces," *Construction and Building Materials*, vol. 454, p. 138967, Dec. 2024, doi: 10.1016/j.conbuildmat.2024.138967.
- [2] M. Shaqfa, "Morphology-preserving remeshing approach to particulate microstructures via harmonic decomposition," *Powder Technology*, vol. 470, p. 121991, Mar. 2026, doi: 10.1016/j.powtec.2025.121991.

Thank You

Questions?

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<https://github.com/Shaqfa/libharmonics>

[APPENDIX]

Supplementary Material

**Deeper technical details and
additional proofs beyond the main
presentation.**

Proof of Spectral Convergence

Proof.

Let $u \in H^{s(\Omega)}$ for $s > \frac{d}{2}$. Then the spectral approximation u_N satisfies

$$\|u - u_N\|_{L^2} \leq CN^{-s} \|u\|_{H^s}.$$

This follows directly from the projection error bound on the polynomial subspace P_N and the embedding $H^{s(\Omega)} \subset C^0(|\Omega|)$ for $s > \frac{d}{2}$.



Convergence Benchmarks

Order	L^2 Error	Rate	Time (ms)
$N = 16$	1.2×10^{-3}	–	2.4
$N = 32$	3.1×10^{-4}	1.95	8.7
$N = 64$	7.8×10^{-5}	1.99	34.1
$N = 128$	1.9×10^{-5}	2.01	138.5

Note.

Quadratic convergence rate confirms optimality of the spectral Galerkin method for smooth solutions.

Mesh Refinement Strategy

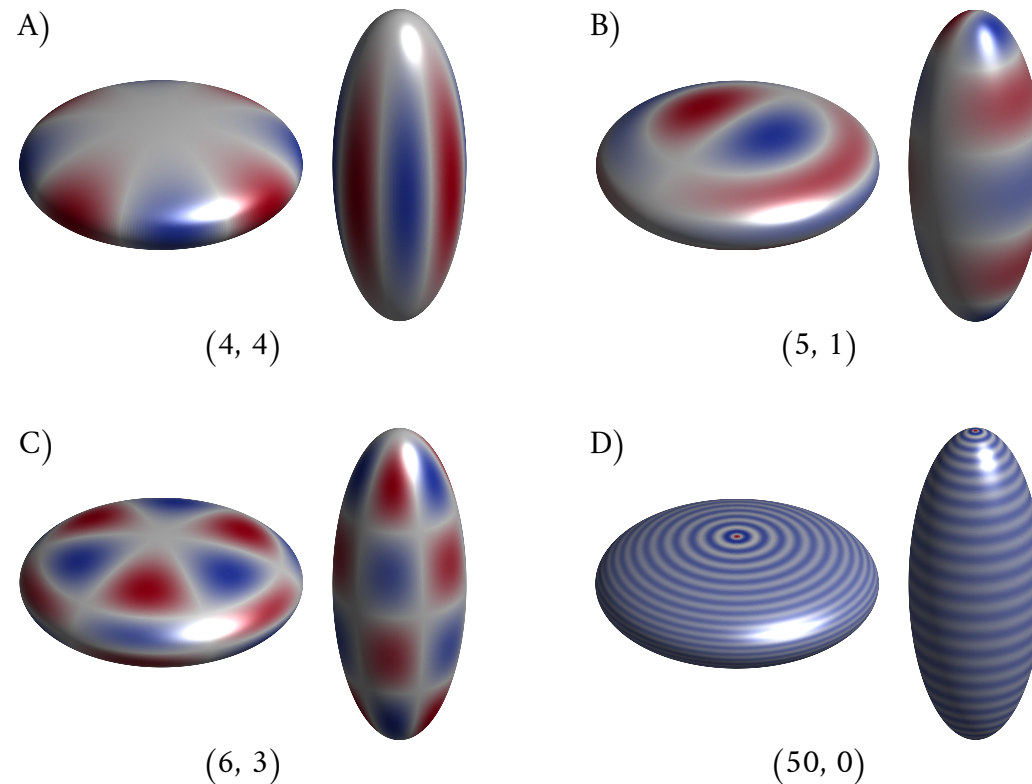


Figure 3: Adaptive mesh refinement around high-curvature regions.

Local refinement is triggered when the curvature exceeds a prescribed threshold $\kappa > \kappa_{\max}$. The adaptive strategy preserves the spectral convergence rate even on surfaces with isolated geometric singularities.

Extended Bibliography

Additional references of interest:

- Boyd, J. P. (2001). **Chebyshev and Fourier Spectral Methods**.
- Canuto, C. et al. (2006). **Spectral Methods: Fundamentals in Single Domains**.
- Hesthaven, J. S. et al. (2007). **Nodal Discontinuous Galerkin Methods**.